

REVIEW

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A Narrative Review of Effects and Underlying Mechanisms of Needling Therapy for Treating Myofascial Pain and Musculoskeletal Disorders

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Abstract

This review aims to summarise the effects and underlying mechanisms of needling therapy (NT) in the treatment of myofascial pain and musculoskeletal (MSK) disorders. This review aims doing so by categorising NT effects according to different anatomic levels of the human body and linking each effect to the specific mechanism. Acupuncture, the first form of NT, has existed for 3000 years and been practised by Traditional Chinese Medicine (TCA); however, its mechanism is hard to be validated by modern science as it involves “Chi”, “Yin & Yang” and “meridians”—all the ambiguous concepts which are hard to be interpreted by the West and modern science. Dry needling (DN), another form of NT, which emerged in the West in the twentieth century, has not well been explicitly reviewed for its mechanisms. Given that NT—acupuncture originated in the East and DN originated in the West—has been proven by systematic reviews and meta-analyses for its effectiveness to treat MSK disorders (world's major disability contributor as defined by World Health Organization) and that many MSK disorders are caused by myofascial pain, a summary of NT effects and mechanisms in modern scientific terminology can provide practitioners with clearer clinical reasoning to justify their treatment using NT to treat myofascial pain and MSK disorders. This review summarised 4 main effects according to different anatomic levels of the human body attributed to NT. They are myofascial effect, supraspinal effect, spinal effect and local effect. The mechanisms responsible for each of these effects, respectively, are suppressing motor end plate activity, descending inhibition mechanism, control gate mechanism and axon reflex.

Keywords Needling therapy, Acupuncture, Dry needling, Myofascial pain, Musculoskeletal disorders, Mechanisms, Effects

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